



Agriculture: Corn Pest Management Guidelines

Charcoal Rot

Macrophomina phaseolini

Symptoms and Signs

Charcoal rot first becomes noticeable when corn is in the tassel stage or later. Infected stalks become shredded; the pith is completely rotted, leaving stringy vascular strands more or less intact. Small, black, spherical sclerotia of the fungus are found on and in the vascular strands; they are numerous enough to give the internal stalk tissue a gray color. As plants mature, the fungus grows into the lower internodes of the stalk, causing the plants to ripen prematurely and weakening the stalks, which may cause them to break.

Comments on the Disease

The pathogen overwinters and is disseminated as sclerotia. Plants are infected through roots only after being predisposed by water stress. The fungus is favored by high temperatures.

Management

Good water management to avoid stressing plants is important in managing this disease, particularly as the crop approaches the flowering stage. Crop rotation to nonhost crops, such as small grains, can also help reduce the disease potential. There are no registered fungicides to control charcoal rot.



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[R.M. Davis\(https://caes.ucdavis.edu/people/mike-davis\)](https://caes.ucdavis.edu/people/mike-davis) (emeritus), Plant Pathology, UC Davis

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